Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

PES1UG24CS058 - Anagha N

Jira space: Scrum_BPS#25 (company-managed Scrum, key SCRUMBPS25)

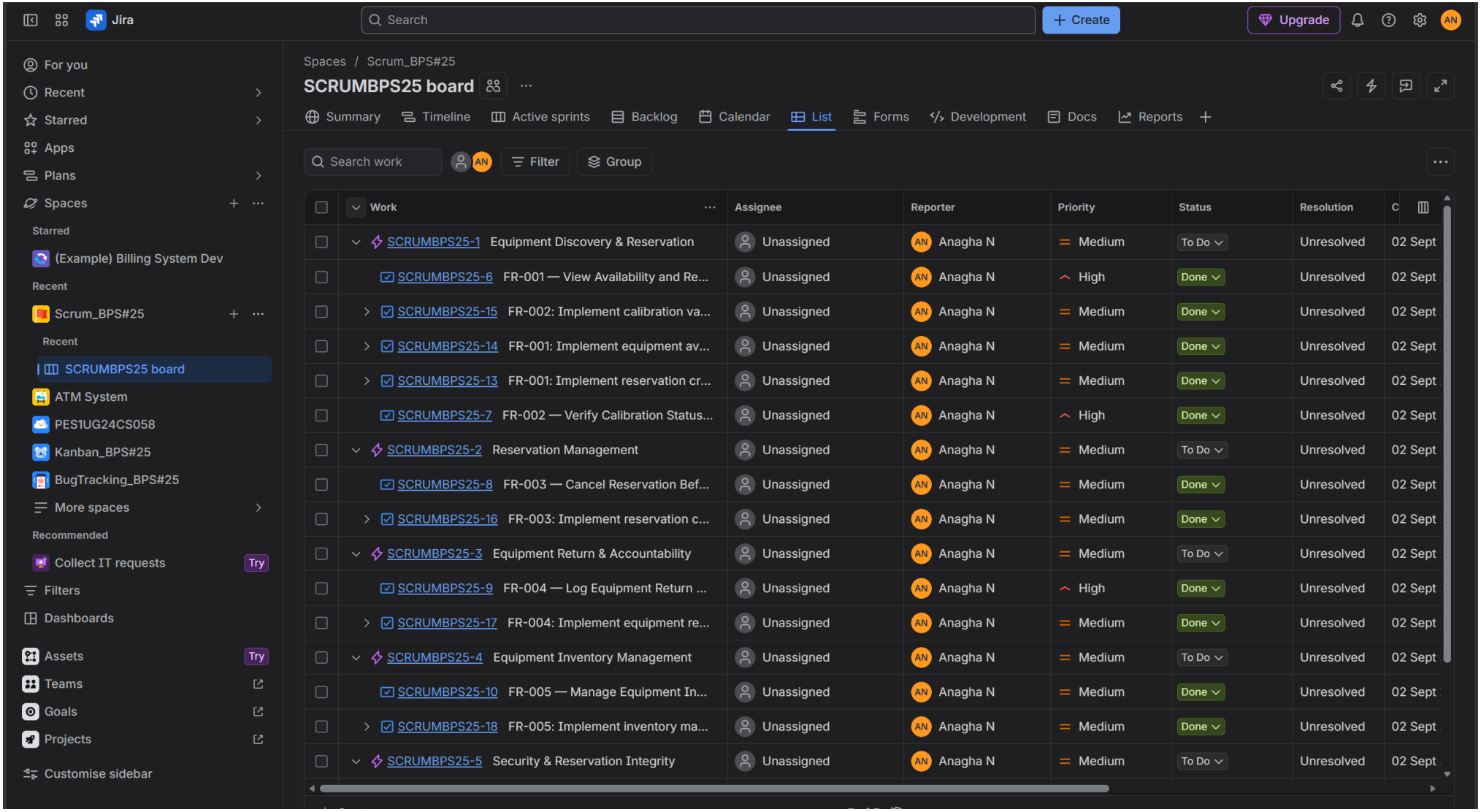
Scenario: Smart Lab Equipment & Slot Reservation Portal (Lab 1 requirements)

Contents:

1. Epics and user stories (backlog)
2. Story point assignments
3. Sprint board (Active Sprint view)
4. Burnup / burndown charts (Sprint 1)
5. Sprint reports (Sprint 1 and Sprint 2)
6. Reflection

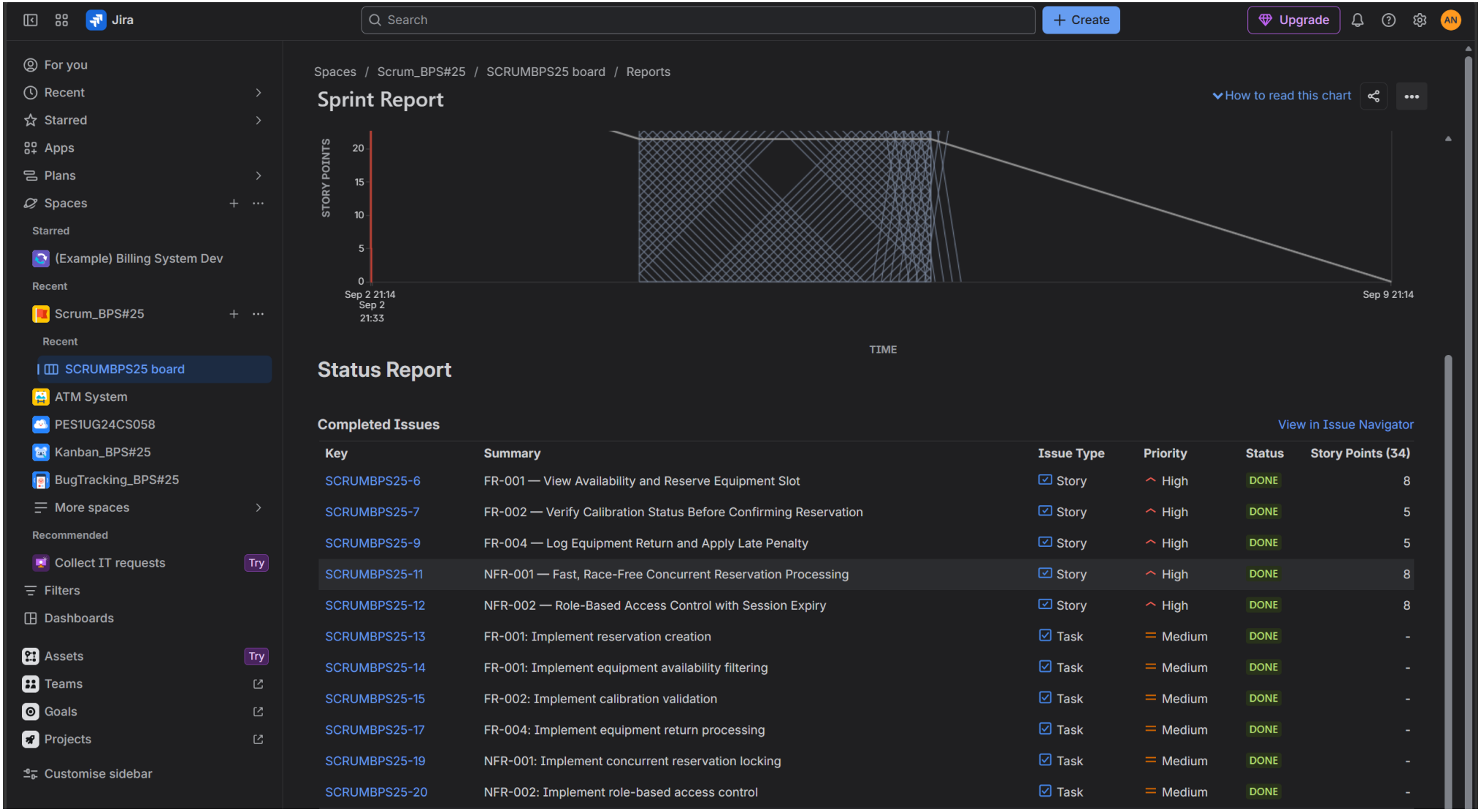
1. Epics and user stories

5 epics (SCRUMBPS25-1 to 5) with their stories and tasks, priority and status



2. Story point assignments

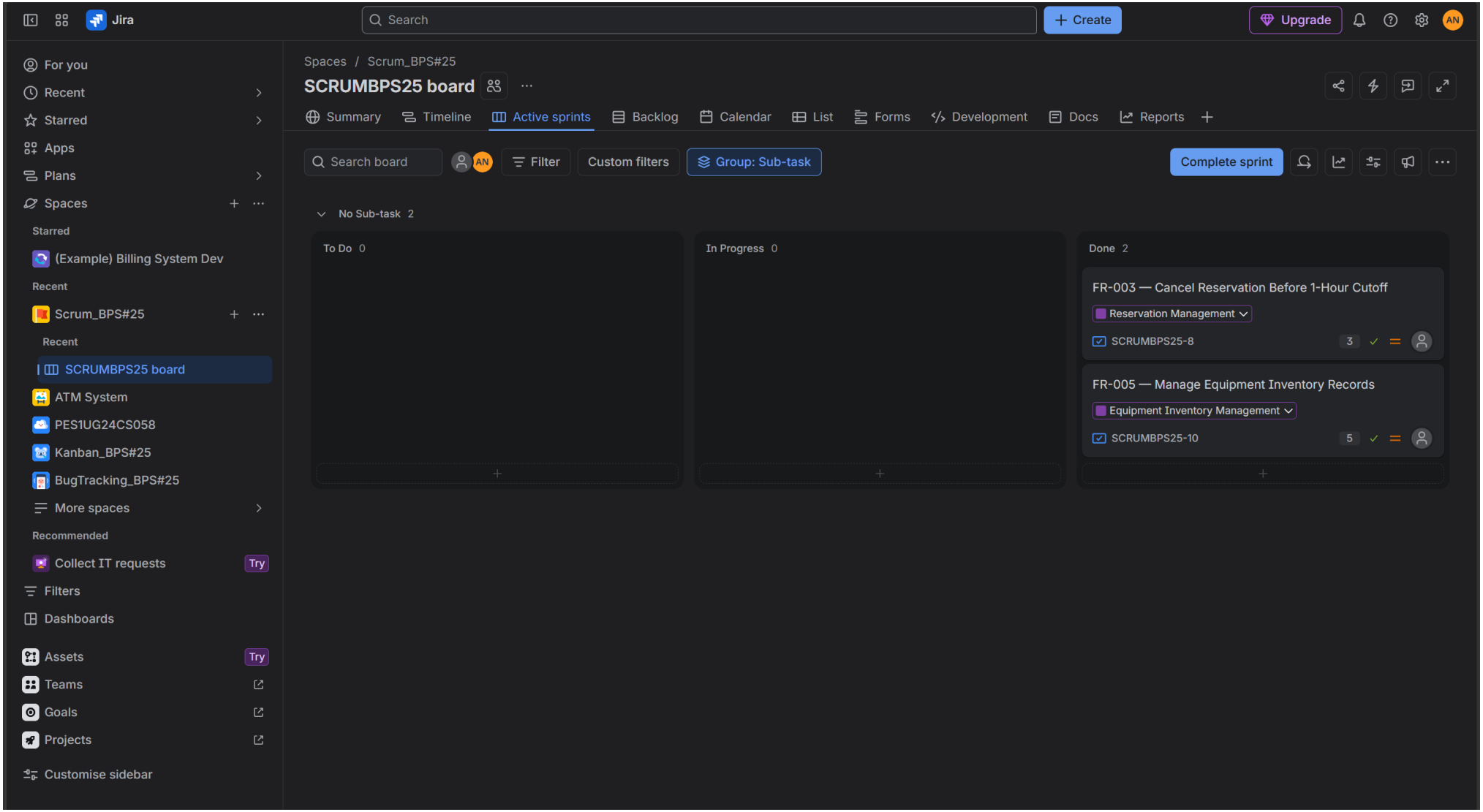
Sprint 1 completed issues, 34 story points (Fibonacci scale)



Key	Summary	Issue Type	Priority	Status	Story Points (34)
SCRUMBPS25-6	FR-001 — View Availability and Reserve Equipment Slot	Story	High	DONE	8
SCRUMBPS25-7	FR-002 — Verify Calibration Status Before Confirming Reservation	Story	High	DONE	5
SCRUMBPS25-9	FR-004 — Log Equipment Return and Apply Late Penalty	Story	High	DONE	5
SCRUMBPS25-11	NFR-001 — Fast, Race-Free Concurrent Reservation Processing	Story	High	DONE	8
SCRUMBPS25-12	NFR-002 — Role-Based Access Control with Session Expiry	Story	High	DONE	8
SCRUMBPS25-13	FR-001: Implement reservation creation	Task	Medium	DONE	-
SCRUMBPS25-14	FR-001: Implement equipment availability filtering	Task	Medium	DONE	-
SCRUMBPS25-15	FR-002: Implement calibration validation	Task	Medium	DONE	-
SCRUMBPS25-17	FR-004: Implement equipment return processing	Task	Medium	DONE	-
SCRUMBPS25-19	NFR-001: Implement concurrent reservation locking	Task	Medium	DONE	-
SCRUMBPS25-20	NFR-002: Implement role-based access control	Task	Medium	DONE	-

3. Sprint board (Active Sprint view)

SCRUMBPS Sprint 3: To Do / In Progress / Done columns, story points on each card



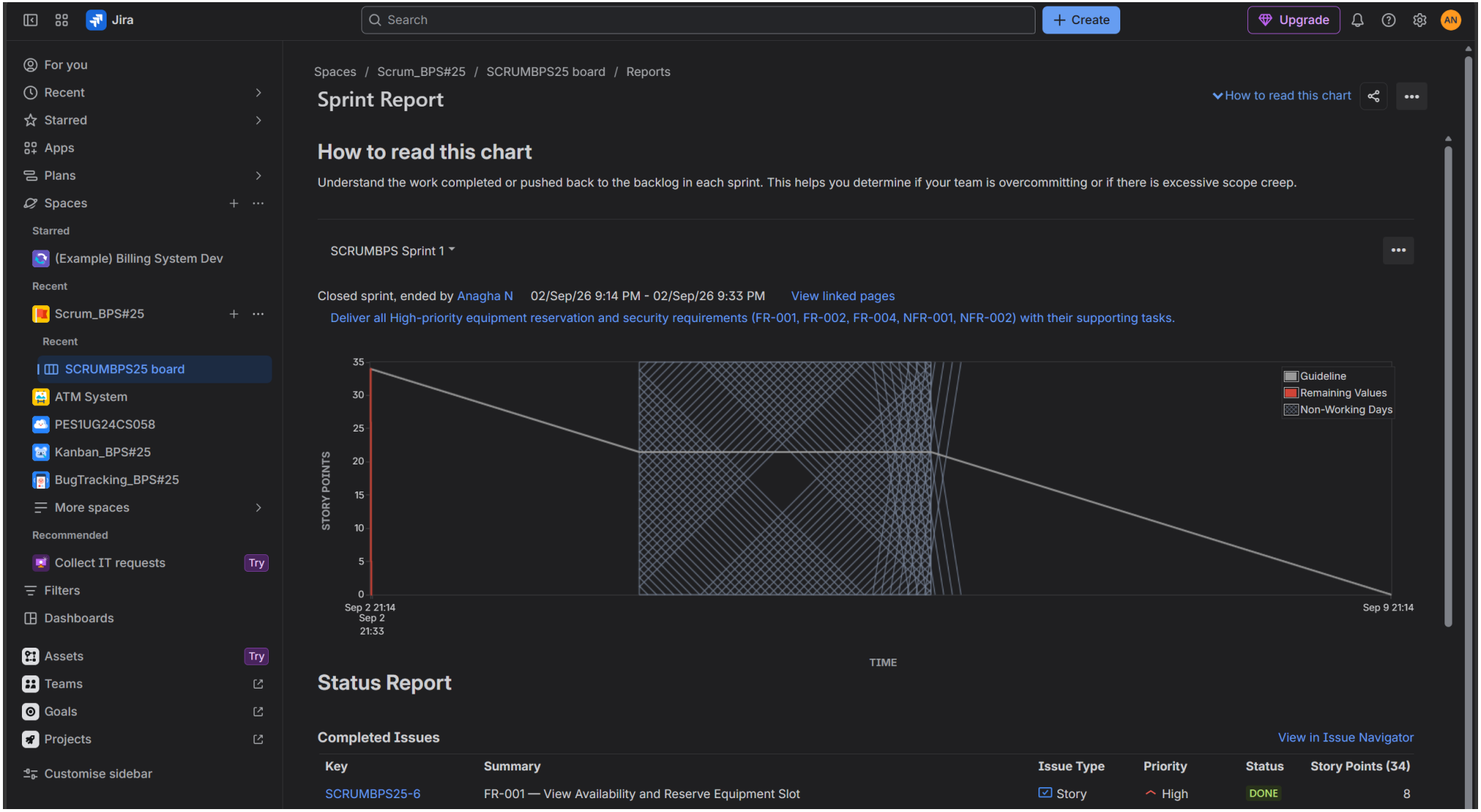
4a. Burnup chart, SCRUMBPS Sprint 1

Reports > Burnup chart: 34 points scope, completed work against guideline



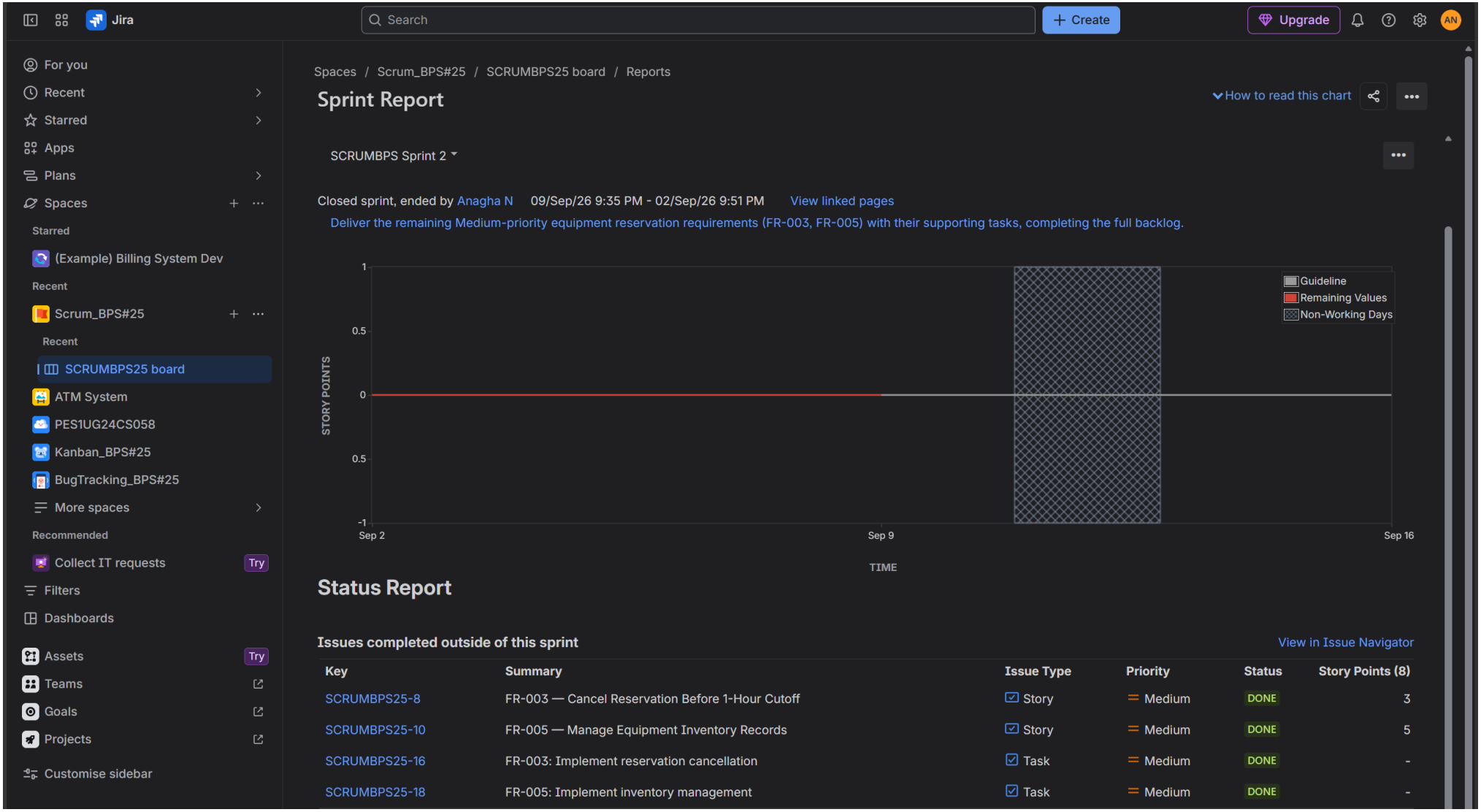
4b. Sprint 1 report, remaining work against guideline

Burndown of remaining story points



5. Sprint 2 report

8 story points (FR-003, FR-005)



6. Reflection

1. Did your estimations reflect the actual effort?
Partly. The 8-point stories (FR-001 reservation, NFR-001 concurrency, NFR-002 access control) are the ones with the most rules to implement, and each has two tasks and sub-tasks. The 3-point story (FR-003 cancellation) was the smallest. Because the sprints were simulated, no real time was measured, so the estimates were not compared against actual effort.

2. Was your backlog well-prioritized?
Yes. The five High-priority stories (FR-001, FR-002, FR-004, NFR-001, NFR-002) went into Sprint 1 and the two Medium-priority ones (FR-003, FR-005) into Sprint 2. Reservation, calibration checks, concurrency and access control are what the system cannot work without. Cancellation and inventory editing depend on them.

3. How did your simulated sprint align with your plan?
Sprint 1 planned 34 points and completed 34 (everything Done). Sprint 2 planned 8 points. Both sprints were run on the same day, so the Sprint 2 stories were finished before its start date and Jira reports them as completed outside the sprint.

4. What insights did the burndown chart give about your team's capacity?
Sprint 1's burnup shows work completed in steps (8, then 16, 24, 29 and 34 points) and ending above the guideline line. Finishing everything in one sitting means the chart shows simulated progress, not real capacity. Sprint 1 carried 34 points against 8 in Sprint 2, so the load was unbalanced. A real team would set Sprint 1 below its demonstrated velocity.